

Acute generalized exanthematous pustulosis: European expert consensus for diagnosis and management

Abstract

Acute generalized exanthematous pustulosis (AGEP) is a rare, usually drug-induced, acute pustular rash. Despite the lack of strong evidence supporting the efficacy of topical or systemic corticosteroids in this condition, they are commonly used. There is currently no consensus on the diagnostic approach or management of AGEP. This paper presents European expert recommendations for the diagnosis and management of AGEP.

The ToxiTEN group of the European Reference Network (ERN)-Skin—comprising dermatologists and allergologists with expertise in drug reactions—developed these recommendations based on clinical experience and a review of the literature. The recommendations are organized into four categories: professionals involved, diagnosis of AGEP, patient management, and allergological evaluation after the acute phase.

Consensus was reached on key aspects including the multidisciplinary team required, minimum diagnostic work-up, treatment setting, therapeutic interventions, and the timing and methods of allergological follow-up. A multidisciplinary approach involving dermatologists, allergologists, and pharmacovigilance services is strongly recommended.

Background and Current Knowledge

Acute generalized exanthematous pustulosis (AGEP) is a rare, typically drug-induced, severe cutaneous adverse reaction characterized by an acute pustular eruption. The estimated incidence ranges from 1 to 5 cases per million per year. The prognosis is generally excellent, with a mortality rate below 5%. AGEP primarily affects adults, with a median age of onset around 60 years, and shows a female predominance.

Clinical Features

AGEP presents with the sudden onset of **widespread erythema**, predominantly in intertriginous areas (e.g., axillae, groin), accompanied by numerous **pinpoint, non-follicular, sterile pustules**. Fever is almost always present. Additional cutaneous manifestations may include:

- Facial edema
- Purpuric lesions
- Atypical target-like lesions
- Vesicles
- Rarely, bullae

Pustules may coalesce, leading to superficial epidermal detachment resembling a **pseudo-Nikolsky sign**. Mucosal involvement occurs in less than 20% of cases and is typically limited to the oral mucosa.

Systemic Involvement

Systemic complications occur in approximately 17% of patients and may involve:

- Liver (most common)
- Kidneys
- Bone marrow
- Lungs (rare)

Hemodynamic instability, including hypotension or shock, is reported in 22% of cases. These features raise the possibility of overlap with other severe cutaneous adverse reactions such as **drug reaction with eosinophilia and systemic symptoms (DRESS)** or, less commonly, **toxic epidermal necrolysis (TEN)**.

Laboratory Findings

Common laboratory abnormalities include:

- **Leukocytosis with neutrophilia**
- **Elevated C-reactive protein (CRP)**
- **Mild eosinophilia** (in 30–50% of cases)

Diagnosis

The **EuroSCAR scoring system** is a validated tool to assess the likelihood of AGEP and differentiate it from other pustular eruptions.

Skin biopsy is essential for confirmation. Histopathology typically shows:

- **Spongiform subcorneal or intraepidermal pustules**
- **Edematous papillary dermis**
- **Perivascular inflammatory infiltrate** rich in neutrophils, with some eosinophils

Occasionally, necrotic keratinocytes or leucocytoclastic vasculitis may be seen.

Differential Diagnosis

The main differential diagnosis is **generalized pustular psoriasis (GPP)**.

Histological clues favoring GPP include:

- Psoriasiform acanthosis
- Papillomatosis
- Tortuous or dilated capillaries

A personal or family history of psoriasis is present in 7–10% of AGEP cases, complicating the distinction between AGEP and GPP.

Special Considerations

- **Localized exanthematous pustulosis (ALEP)** has been described, particularly with topical medications, though its pathogenesis remains unclear.
- Comorbidities such as diabetes, renal failure, prior psoriasis, and history of hypersensitivity reactions are frequently reported in AGEP patients.
- AGEP patients may have a higher body mass index compared to those with DRESS.

A multidisciplinary approach is essential for accurate diagnosis, appropriate management, and pharmacovigilance reporting.